Tingxi Li

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Research Interests

I am broadly interested in the **inference-time efficiency** of deep learning systems across two levels: (i) kernel and subgraph-level efficiency, and (ii) model-level dynamicity. My primary research interests lie in **deep learning compilation**, targeting compiler infrastructure and AI-driven kernel generation. I also investigate dynamic behavior vulnerabilities in deep learning models and develop defenses against efficiency exploitation.

Publications

Characterizing Real-World Bugs in Tile Programs for Automated Bug Detection ***(Distinguished Paper)***
R. Rathnasuriya, Z. Song, N. Majoju, ***T. Li****, A. Moharir, W. Yang, T. Xie* — *ISSTA'26*

SoK: Efficiency Robustness of Dynamic Deep Learning Systems
R. Rathnasuriya, ***T. Li****, Z. Xu, Z. Song, M. Haque, S. Chen, W. Yang* — *USENIX Sec'25*

COMET: Closed-loop Orchestration for Malicious Elicitation Techniques in Code Models
Z. Xu, ***T. Li****, R. Rathnasuriya, Z. Song, J. Ren, B. Mandalapu, S. Setayeshpour, X. Du, W. Yang* — *Technical Report*

An Empirical Study of GPU Kernel Performance Gaps in Modern Domain-Specific Languages
T. Li*, R. Rathnasuriya, W. Yang* — *Under Review*

AESOP: Adversarial Execution-path Selection to Overload Deep Learning Pipelines
T. Li*, M. Ji, R. Rathnasuriya, S. Chen, Y. Hu, W. Yang* — *arXiv Preprint*

Research Projects

A Survey of Automation in Kernel Generalization and Optimization **Jan. 2026 – Present**

- Systematized DL compilation research into AI-driven kernel generation and compiler infrastructure, with subcategories on search space pruning, auto-tuning, cost modeling, and candidate validation.
- Analyzed IR evolution from loop-based to tile-based abstractions and its downstream impact on the compilation ecosystem.

Amazon Trusted AI Competition **Nov. 2024 – Jul. 2025**

- Finalist team ($250K prize) targeting black-box jailbreaking of CodeLLMs.
- Fine-tuned surrogate models for adversarial prompt guidance, achieving 0.80+ recall and 0.75+ F1 in malicious intent detection.

Education

University of Texas at Dallas **Sep. 2024 – Present**
Doctor of Philosophy

Dalian University of Technology **Sep. 2019 – May 2024**
Bachelor of Science

Internship Experience

Sophgo **Jun. 2024 – Aug. 2024**
Research Intern *Shenzhen, China*

- Developed C++ API code for models on RISC-V processors and authored deployment documentation.
- Fine-tuned models on private datasets, identified failure cases, and applied data augmentation to improve robustness.

Teaching

CS 4365 – Artificial Intelligence **Spring 2026**
Teaching Assistant, UT Dallas

CS 4375 – Introduction to Machine Learning **Fall 2025; Fall 2024**
Teaching Assistant, UT Dallas

CS 4337 – Programming Language Paradigms **Spring 2025**
Teaching Assistant, UT Dallas

Miscellaneous

Tech Stack: Python; C; C++; Java; PyTorch; LaTeX; SQL